Data-Driven Business Optimization in Specialty Coffee Roasting: A Case Study of an Afficionado Coffee Roaster

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Abstract

This research presents a comprehensive data-driven evaluation of an Afficionado Coffee Roaster business model, emphasizing revenue analysis, category-wise performance, and the application of business intelligence tools to support strategic decision-making. The study begins with a robust data engineering process, where raw transactional data was cleaned, structured, and transformed to ensure analytical accuracy. Through this process, the total revenue was calculated to be approximately $698,812, forming the foundation for subsequent analysis. Exploratory data analysis was conducted to identify key trends and patterns within the dataset, revealing that coffee is the dominant revenue-generating category, significantly outperforming other segments such as tea and complementary products.

To effectively communicate these insights, an interactive dashboard was designed and enhanced with advanced visual and analytical components. Key upgrades included the addition of a "Total Revenue" KPI card to provide an at-a-glance performance metric, and a pie chart visualization illustrating the revenue share across different product categories, enabling clear comparative analysis. Furthermore, interactive distribution widgets were incorporated to allow dynamic exploration of sales patterns, customer behavior, and category-level contributions. The analytical solution was deployed as a live application using an ngrok tunnel, ensuring real-time accessibility and seamless integration of backend processing with frontend visualization. This deployment approach enabled continuous monitoring and usability of the dashboard in a dynamic business environment. Overall, the study demonstrates how the integration of data engineering, exploratory analysis, and interactive visualization can generate actionable insights, enhance operational efficiency, and support data-driven decision-making in a specialty coffee roasting business.

1 Introduction

The specialty coffee industry has witnessed significant growth in recent years, driven by increasing consumer awareness, evolving taste preferences, and the rising demand for premium and artisanal products. Within this context, Afficionado Coffee Roasters represent a niche yet rapidly expanding segment that emphasizes quality, customization, and customer experience. However, as competition intensifies and product portfolios diversify to include categories such as coffee, tea, and related beverages, businesses must adopt data-driven approaches to remain competitive and sustainable. This study focuses on leveraging data analytics to understand revenue dynamics, product performance, and operational efficiency within an Afficionado Coffee Roaster business model.

The primary objective of this research is to analyze transactional data and transform it into actionable insights that can support strategic decision-making. To achieve this, a comprehensive data engineering process was undertaken, where raw transaction data was processed, cleaned, and structured to ensure accuracy and reliability. Through this process, the total revenue was calculated to be approximately $698,812, providing a quantitative foundation for further analysis. This revenue figure not only reflects the overall business performance but also serves as a benchmark for evaluating category-wise contributions and growth opportunities.

Building upon the processed dataset, exploratory data analysis was conducted to identify underlying patterns and trends. One of the key findings from this analysis was the dominance of the coffee category, which emerged as the primary driver of revenue compared to other segments such as tea and complementary products. To better understand these dynamics, revenue contributions across all categories were mapped and visualized, enabling a clear comparison of their relative impact on total sales. Interactive distribution widgets were also developed to facilitate deeper exploration of sales behavior, allowing users to examine variations in product demand and performance.

To effectively present these insights, a comprehensive and interactive dashboard was developed using Streamlit. The dashboard integrates multiple analytical components, including dynamic filtering options that allow users to narrow down the analysis by product category. Key performance indicators (KPIs), such as Total Revenue, Most Popular Product, and Units Sold, are prominently displayed to provide immediate visibility into business performance. Additionally, comparative visualizations, including side-by-side rankings of volume versus financial performance and revenue share pie charts, enable a more nuanced understanding of the relationship between sales quantity and revenue generation.

Further enhancements were made to the dashboard to improve its analytical depth and usability. A dedicated "Total Revenue" KPI card was introduced to highlight overall performance, while a pie chart visualization was integrated to clearly depict the revenue share of each category, such as coffee and tea. These visual elements not only enhance interpretability but also support data-driven storytelling, making complex information more accessible to stakeholders.

To ensure real-world applicability and accessibility, the dashboard was deployed as a live application. The updated application (app.py) is hosted using an ngrok tunnel, allowing it to run in the background and be accessed beyond the local development environment. This deployment approach enables real-time interaction with the data and ensures that insights can be continuously monitored and utilized for decision-making.

In summary, this research establishes the importance of integrating data engineering, exploratory analysis, and interactive visualization in understanding and optimizing the performance of a coffee roasting business. By combining robust analytical techniques with practical deployment strategies, the study demonstrates how data can be transformed into a strategic asset, ultimately enhancing business efficiency, customer understanding, and revenue growth.

2 Objective

The objectives of this research are as follows:

1. To compute and analyze total revenue from transactional data
2. To perform category-wise and product-type-wise revenue analysis
3. To identify top-performing products based on quantity and revenue
4. To analyze customer purchasing patterns based on time of day

5. To develop an interactive dashboard for data visualization and analysis
6. To enable location-based insights using dynamic filtering

3 Data Description

The dataset used in this study consists of 149,116 transaction records, representing a comprehensive log of sales activities. Each record includes multiple attributes that support multi-dimensional analysis.
Key attributes include:

1. Transaction ID – Unique identifier for each transaction
2. Transaction Time – Timestamp of purchase
3. Transaction Quantity – Number of units purchased
4. Unit Price – Price per unit of product
5. Store Location – Physical store where the transaction occurred
6. Product Category – High-level classification (e.g., Coffee, Tea)
7. Product Type – Detailed classification of products

The dataset enables analysis across multiple dimensions, including product hierarchy, time, and location. Its large size ensures statistical reliability and supports meaningful insights into customer behavior and business performance.

4 Data Engineering

The data engineering process formed the foundation of this study by transforming raw transactional data into a structured format suitable for analysis. The dataset, consisting of 149,116 records, was first cleaned to ensure consistency and accuracy across key attributes such as product category, product type, transaction time, quantity ordered, and unit price. A crucial step in this process was the computation of total revenue at the transaction level using the formula:

$$\mathbf{Revenue} = \mathbf{UnitPrice} * \mathbf{QuantityOrdered}.$$

This transformation enabled the aggregation of revenue across multiple dimensions, including category, product type, and store location, resulting in an overall revenue of approximately $698,812. Following data preparation, several analytical methods were applied to extract meaningful insights. Category-wise revenue analysis was conducted to understand the contribution of each product segment, supported by pie chart visualizations. Product-type distribution was analyzed to examine internal category composition, while comparative analysis was performed by ranking products based on both quantity sold and revenue generated to identify discrepancies between high-volume and high-value items. Additionally, revenue share analysis highlighted the dominance of coffee (38.6%) followed by tea (28%). Time-based analysis was carried out by segmenting transaction data into morning, noon, and evening periods, revealing distinct customer purchasing patterns. These engineered datasets and analytical approaches enabled efficient visualization and supported the development of an interactive dashboard for dynamic business insights.

5 Methodology

The methodology adopted in this study follows a structured analytical approach, combining quantitative analysis techniques with interactive visualization to derive meaningful business insights from transactional data. The process begins with revenue analysis, where total revenue is computed at the transaction level and aggregated across categories and product types to establish overall business performance. This is followed by category-wise analysis, where the contribution of each product category is evaluated using proportional visualization techniques such as pie charts to clearly represent revenue distribution.

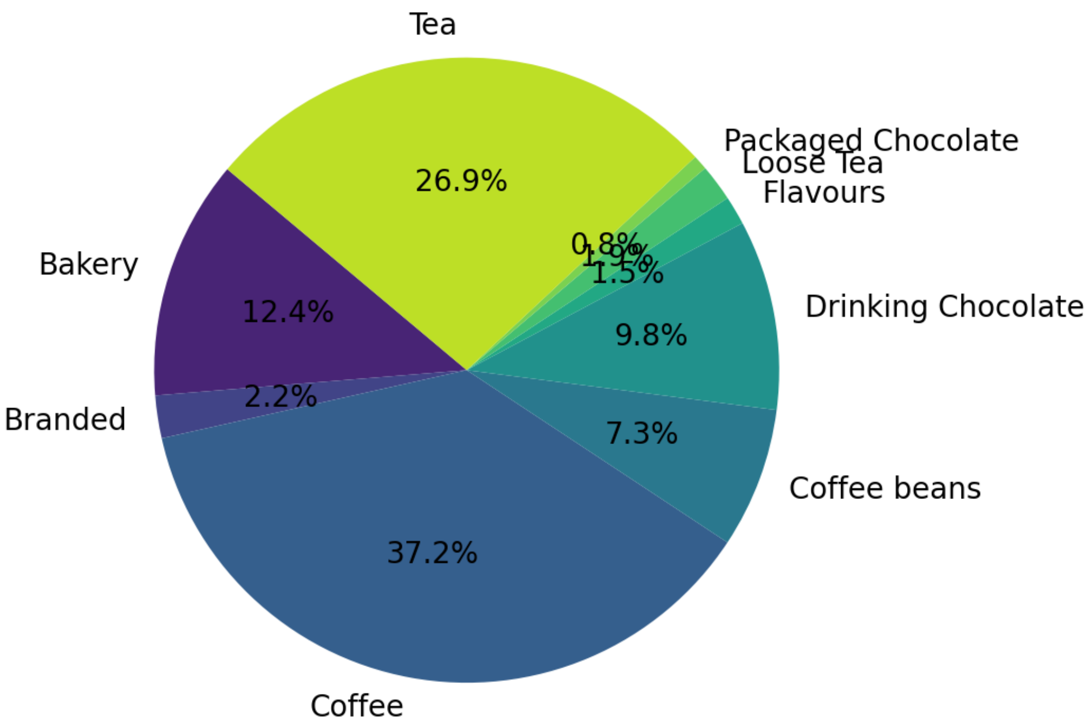

Figure 1: Revenue share: Type-wise

To further understand product-level dynamics, a product-type distribution analysis is conducted using pie chart visualizations, enabling identification of dominant product types within each category. A key analytical technique employed is comparative ranking analysis, where products are ranked based on two critical metrics—quantity sold and revenue generated. This dual-ranking approach helps in distinguishing between high-demand products and high-profit products, thereby revealing potential gaps in pricing or sales strategy.This involves examining how individual products contribute within their respective categories. For instance, within the coffee category, high-frequency items such as brewed coffee and espresso-based beverages show strong distribution, indicating consistent customer demand. Similarly, tea variants contribute significantly within their category, reflecting consumer preference diversification.Refer Figure2

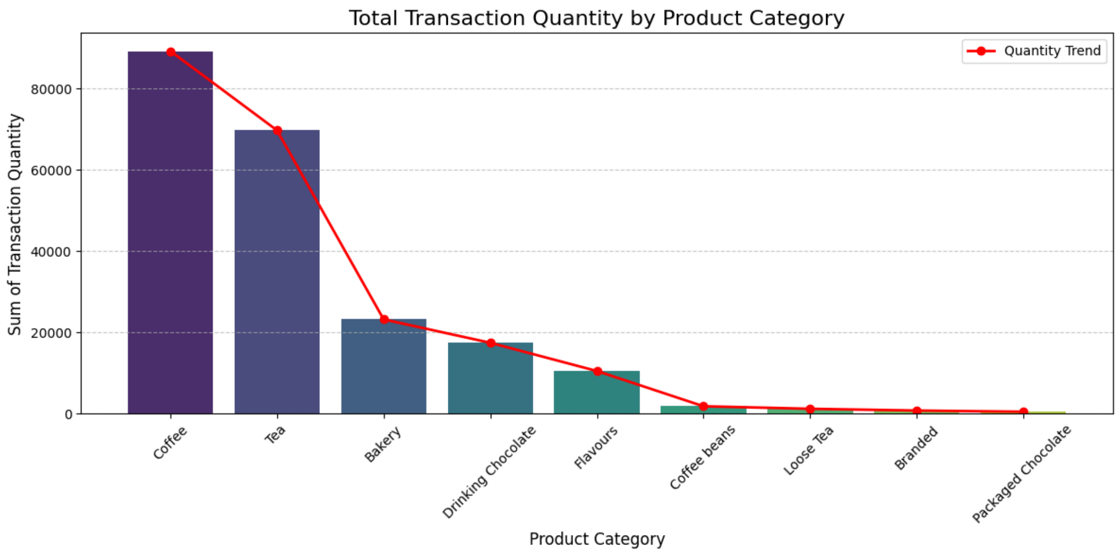

Figure 2: Product ranking: By Order

Additionally, a top-N analysis is performed to identify the top 10 products based on quantity sold, highlighting the "hero product" that drives the highest demand. To capture customer behavior patterns, a time-series segmentation technique is applied by categorizing transaction timestamps into three distinct periods: morning, noon, and evening. This enables the analysis of temporal purchasing trends, providing insights into peak demand periods for different product categories.This allows the identification of a "hero product," typically a frequently purchased item such as brewed coffee, which plays a central role in driving daily sales volume. This analysis is visualized using bar charts for clear comparison.Refer Figure 3

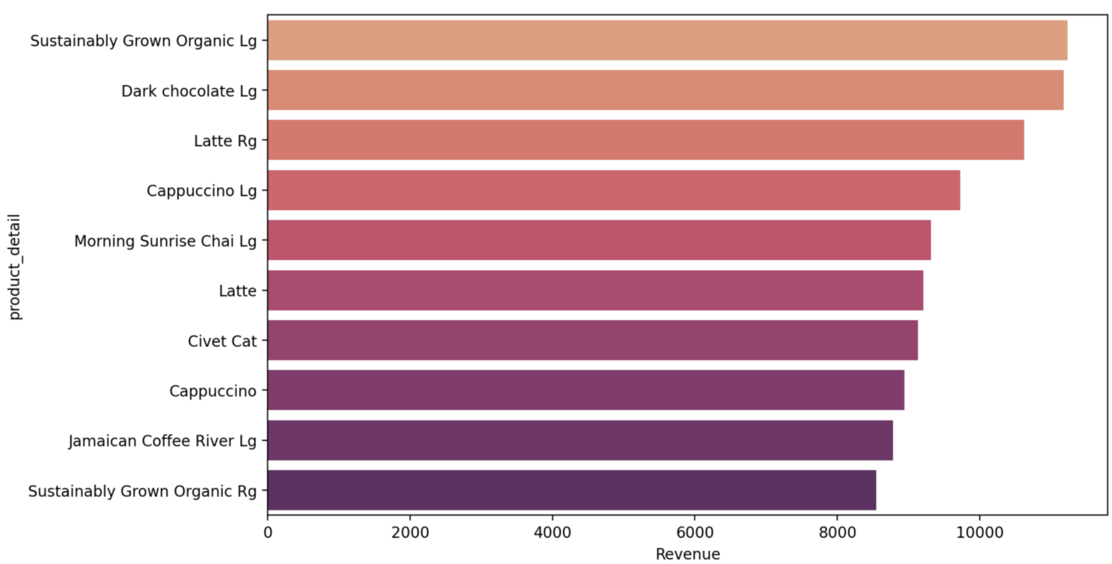

Figure 3: Top Product:By Popularity

A key analytical approach used is comparative ranking analysis, where products are ranked based on two dimensions: total quantity sold and total revenue generated. This technique reveals important business insights. For example, products like brewed coffee may rank highest in terms of quantity sold, indicating high demand, while premium items such as latte or cappuccino may generate higher revenue despite relatively lower sales volume. This distinction helps identify "volume drivers" versus "revenue drivers" within the product portfolio.Refer Fig 4.

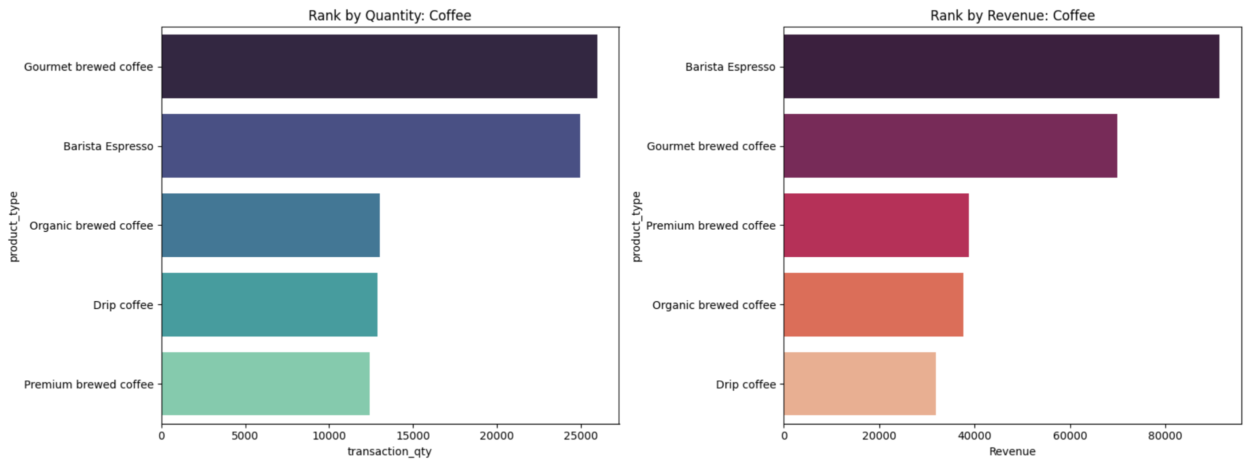

Figure 4: Compare: Revenue vs Qty Order

To understand customer behavior patterns, a time-based segmentation analysis is applied using transaction timestamps. The dataset is divided into three time periods: morning, noon, and evening. This reveals distinct consumption patterns, where coffee products such as espresso, cappuccino, and brewed coffee are predominantly ordered during morning hours, while tea products gain higher preference during evening periods. This temporal trend reflects real-world consumer habits and supports targeted business strategies.

Additionally, a revenue share analysis at the item-type level is conducted, quantifying the percentage contribution of each category. The findings indicate that coffee contributes approximately 38.6% of total revenue, while tea accounts for around 28%, reinforcing the dominance of coffee while also highlighting the importance of tea as a secondary category.

Finally, the methodology incorporates interactive data exploration techniques through a Streamlit-based dashboard. Features such as sidebar filtering allow users to analyze data based on store location, enabling localized insights into product performance and customer preferences. This enhances the practical applicability of the analysis by allowing dynamic, real-time interaction with the data.Refer Fig 5

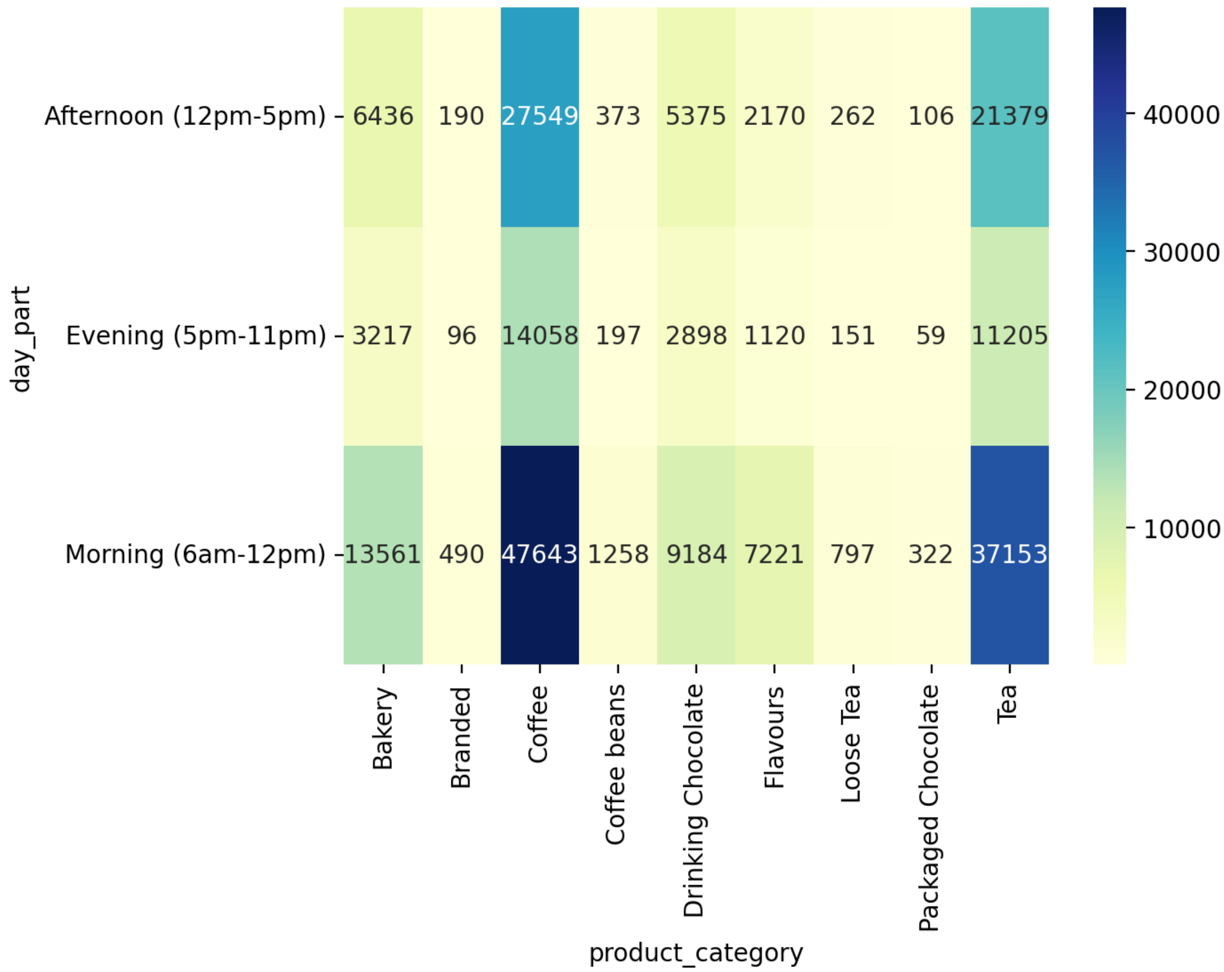

Figure 5: Behavior Analysis: Time wise

Through the implementation of location-based filtering in the dashboard, variations in product performance across store locations were observed. While coffee remains the dominant category across all locations, the proportion of tea and other products varies slightly, indicating localized preferences. Location-Based Revenue Analysis

The location-wise revenue analysis reveals significant variation in sales performance across different store locations. Among the three analyzed locations, Hell's Kitchen emerges as the highest revenue-generating store, with a total revenue of approximately $236,511.17, followed by Astoria with $232,243.91 and Lower Manhattan with $230,057.25. Although the difference in revenue across locations is relatively moderate, Hell's Kitchen consistently outperforms the others, indicating a slightly higher concentration of customer activity and purchasing behavior.

This performance can be attributed to factors such as higher foot traffic, favorable location dynamics, or stronger demand for key product categories like coffee and espresso-based beverages. The relatively close revenue figures across all three locations also suggest a well-balanced distribution of business operations, reducing dependency on a single location. However, the leading position of

Hell's Kitchen highlights its strategic importance and presents an opportunity for targeted marketing, optimized inventory allocation, and potential expansion of high-performing product offerings in this area.

Figure 6: Revenue Distribution Across Store Locations

6 Dashboard Overview and Interpretation

The developed dashboard serves as an interactive decision-support tool that enables users to explore and interpret business performance in a comprehensive and intuitive manner. At the top level, the dashboard presents key performance indicators (KPIs), including Total Revenue $(698, 812), TotalUnitsSold(214, 470), AverageOrderValue(4.69)$, and the Most Popular Product (Earl Grey Rg), providing an immediate snapshot of overall business health. These metrics allow users to quickly assess performance without needing to navigate deeper into the dataset.

The dashboard further facilitates comparative analysis through dedicated visual sections, such as Tea vs Coffee comparison charts, where both quantity and revenue perspectives are presented side-by-side. This enables users to understand not only which category sells more but also which generates higher financial returns. Additionally, the Revenue Trend (Moving Average) visualization helps users identify patterns over time, smoothing short-term fluctuations to reveal underlying trends in business performance.

A key strength of the dashboard lies in its interactivity. The sidebar filters, including store location and time-of-day selection, allow users to dynamically adjust the analysis based on specific criteria. For example, users can isolate performance for a particular store or examine purchasing behavior during different time periods such as morning, noon, or evening. This feature enhances the ability to derive localized and time-specific insights.

The Product Type Deep Dive section provides granular analysis by allowing users to focus on a selected category, such as coffee, and examine its internal distribution through pie charts and quantity-based visualizations. This helps in identifying dominant product types within a category. Furthermore, the Top 5 Hero Products visualization highlights the highest-performing items, enabling quick identification of key revenue and volume drivers.

The inclusion of a demand heatmap offers an additional layer of insight by visually representing peak purchasing periods, helping users understand customer behavior patterns at different times. Overall, the dashboard integrates KPIs, comparative visuals, trend analysis, and interactive filtering into a unified interface, allowing users to seamlessly navigate from high-level summaries to detailed insights. This structured design ensures that both technical and non-technical users can effectively interpret the data and make informed, data-driven decisions.Ref Fig7

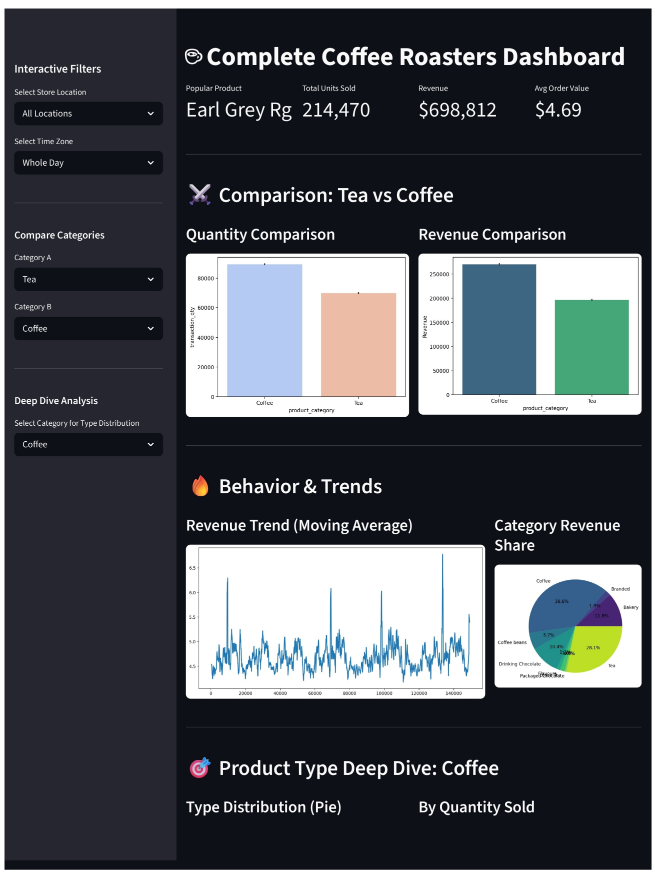

Figure 7: Dashboard Preview

7 Menu Diversification Recommendation

Based on the analysis of product popularity and revenue contribution, menu diversification can be strategically implemented to enhance both profitability and customer satisfaction. High-demand items such as brewed coffee (volume driver) and high-revenue products like latte and cappuccino indicate strong customer preference for coffee-based beverages. Similarly, the popularity of tea variants such as Earl Grey highlights a consistent secondary demand. To capitalize on these insights, the business can introduce variations of top-performing products, such as flavored lattes, seasonal brews, or premium tea blends, thereby expanding the menu without deviating from customer preferences. Additionally, bundling popular items or offering time-specific combos (e.g., morning coffee deals or evening tea specials) can further increase sales. This approach ensures that menu expansion is data-driven, focusing on proven products, ultimately improving customer engagement while maximizing revenue potential.

8 Conclusion

This study demonstrates how data analytics can effectively transform large-scale transactional data into meaningful business insights for an Afficionado Coffee Roaster. By analyzing 149,116 transactions, key patterns in revenue distribution, product performance, and customer behavior were identified, highlighting coffee as the primary revenue driver and revealing distinct time-based consumption trends. The development of an interactive dashboard further enhanced the accessibility and practical application of these insights, enabling dynamic exploration and informed decision-making. Overall, the research emphasizes the importance of a data-driven approach in optimizing product strategy, improving customer experience, and supporting sustainable business growth.

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